Southern Cross University

Pharmacology of the Cardiovascular System

Thrombosis

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COMMONWEALTH OF AUSTRALIA

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Objectives for this portion of the chapter

- **Explain the various steps involved in thrombus formation**
- **Explain how each of our medications that decrease thrombus formation affect one of the steps involved in thrombus formation,**
- **Identify and explain the actions of common pharmacological agents used to prevent the development of a thrombus.**



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Medications of the Cardiovascular System: Thrombosis

Lesson 1:

Understanding the foundational information about haemostasis

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Thrombus

Plural form = thrombi
Process of thrombus formation = thrombosis

A thrombus is a blood clot

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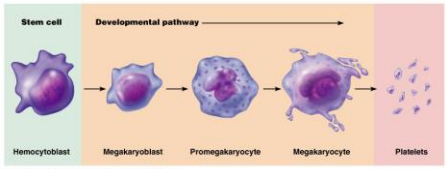
Haemostasis

The process of haemostasis involves:

1. **Vessel spasm**
2. **Formation of the platelet plug**
3. **Blood coagulation or development of an insoluble fibrin clot**
4. **Clot retraction**
5. **Clot dissolution**

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Differentiation of precursor cells to platelets

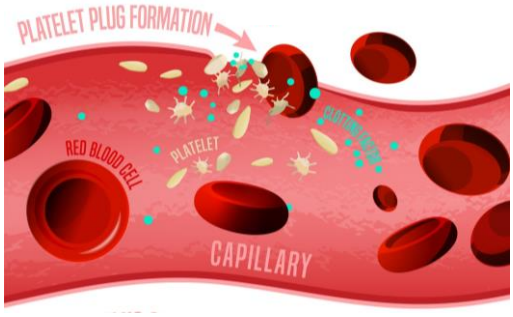


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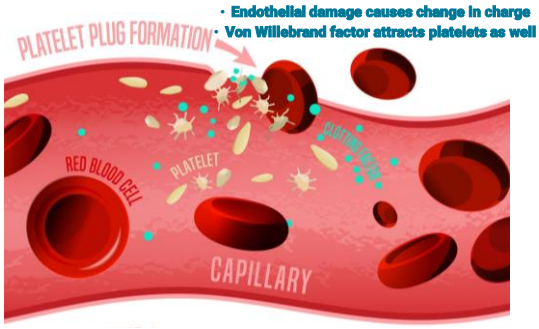
Platelets contain:

- Contractile proteins including actin and myosin
- Mitochondria that produce ADP
- Preformed enzymes that synthesise prostaglandins
  - Cyclooxygenase (COX) synthesises pro-platelet prostaglandin TXA2; thromboxane A2 (intact endothelium contains PGI2; prostacyclin)
- Growth factors that repair damage to endothelium

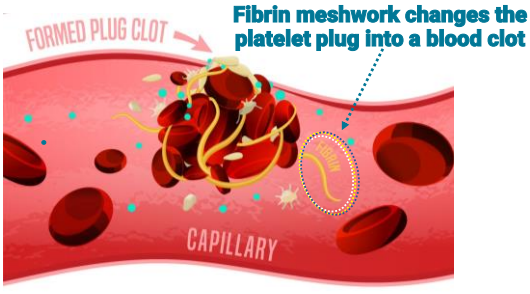
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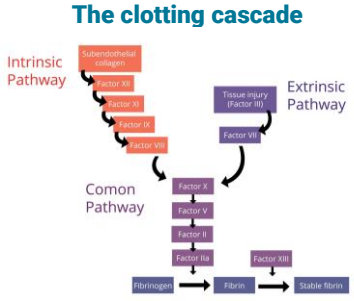
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PLATELET PLUG FORMATION
• Endothelial damage causes change in charge
• Von Willebrand factor attracts platelets as well
RED BLOOD CELL
PLATELET
CAPILLARY

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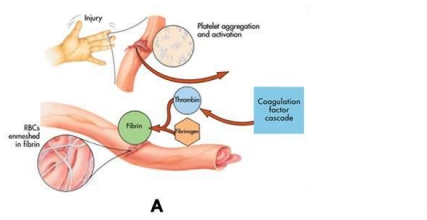

FORMED PLUG CLOT
Fibrin meshwork changes the platelet plug into a blood clot
CAPILLARY

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The clotting cascade
Intrinsic Pathway
Subendothelial collagen
Factor XII
Factor XI
Factor IX
Factor VIII
Extrinsic Pathway
Tissue injury (Factor III)
Factor VII
Comon Pathway
Factor X
Factor V
Factor II
Factor IIa
Factor XIII
Fibrinogen
Fibrin
Stable fibrin

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The clotting cascade

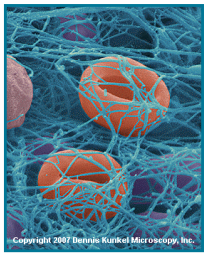


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(A from Thibodeau GA, Patton KT: *Anatomy & physiology*, ed 5, St Louis, 2003, Mosby;

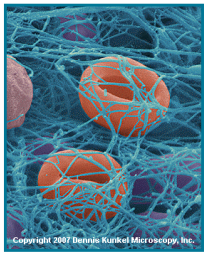
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Fibrin



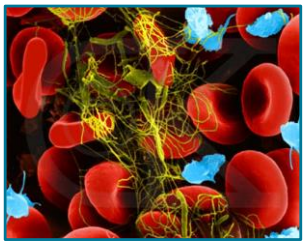
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Ultimately, plasminogen changes to plasmin which degrades the clot



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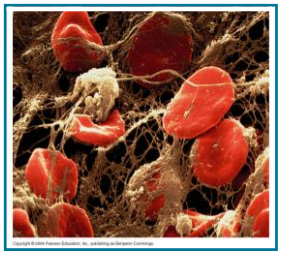
Arterial Clot



http://www.treehugger.com/files/2007/09/how_air_pollution.php

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Venous Clot



Pearson Education

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Medications of the Cardiovascular System:
Thrombosis
Lesson 2:
Antiplatelet drugs

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Antiplatelet drugs: Aspirin
aka acetylsalicylic acid (ASA)

- Aspirin used long-term to help prevent further heart attacks, ischaemic strokes, and arterial blood clots in people at high risk.
- Aspirin works similarly to other NSAIDs but also affects platelets
- The bark of the willow tree containing ASA has been used for its health effects for at least 2,400 years.
- Clopidogrel is taking over as preferred antiplatelet drug
  - Can be combined with aspirin

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Antiplatelet drugs: Aspirin

- PGI2 from intact epithelium and TXA2 from platelets normally balance to form ideal haemostasis
  - This balance is disrupted when the intact epithelium is covered in atherosclerotic plaques, making the pro-platelet TXA2 dominate
- The acetyl group in aspirin non competitively inhibits cyclooxygenase (COX), needed to form TXA2.
- Thromboxane (TXA2) is therefore not able to be made for the entire 9 day life of the platelet.
  - Balance is restored
  - Intact endothelium contains PGI2; prostacyclin but the endothelial cells can synthesise more enzyme while platelets cannot

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Aspirin as an antiplatelet drug

- Low dose aspirin (75-150 mg/day) is proven effective in reducing MI and strokes from atherosclerosis
- Adverse effects
  - Stomach ache, stomach ulcers, allergies

Q: I am healthy, and my arteries should be clean, why don't I bleed readily when I take aspirin?

- A: Aspirin doesn't destroy platelets, it takes away one of their 'tools'. Also, not all platelets are affected.

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Clopidogrel

- Clopidogrel is emerging as the preferred antiplatelet medication
- Decreases risk of heart attack and stroke
- Its effect starts about two hours after intake and lasts for five days.
- Common side effects include headache, nausea, easy bruising, itching, and heartburn.
- Rarely causes thrombotic thrombocytopenic purpura.
- Works by irreversibly inhibiting a receptor called P2Y12 for ADP on platelets.

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Medications of the Cardiovascular System:
Thrombosis
Lesson 3:
Anticoagulants

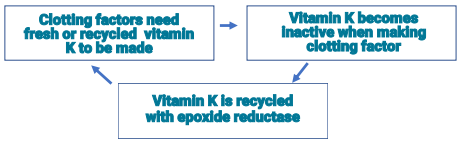
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Anticoagulants

- Anticoagulants, commonly known as blood thinners, but do not affect the viscosity of blood
- Prevent or reduce coagulation of blood, prolonging the clotting time.
- Used for both arterial and venous clots
- One of the first anticoagulants, warfarin
  - Component from sweet clover
    - First developed as a poison: rat bait

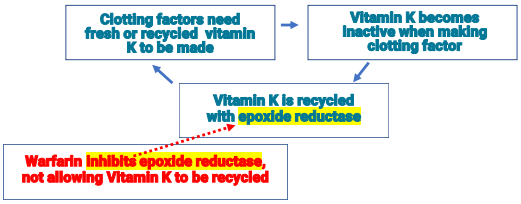
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Simplistic look at the mechanism by which warfarin works



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Simplistic look at the mechanism by which warfarin works



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Drugs that affect coagulation

Warfarin

- Taken orally, absorbed quickly
- Takes several days to exert its effect (there is still active vitamin K hanging around)
- Affects several clotting factors

- Warfarin inhibits epoxide reductase, an enzyme that recycles vitamin K back to its active form
- Without active vitamin K, many coagulation factors are missing, resulting in decreased coagulation and if excessive, bleeding

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Drugs that affect coagulation

Warfarin (continued)

- Dose is extremely variable (2-20mg/day) and requires frequent assessment of coagulation (INR tests need to be performed to make sure of ideal clotting)
- Huge number of drugs, dietary factors affect its efficacy
- Side effects
  - Allergy, bruising, excessive bleeding
  - WARFARIN IS TEROTAGENIC
    - (contraindicated in women of child bearing age)

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Drugs that affect coagulation

Rivaroxaban

- An oral anticoagulant
- Has fewer side effects than warfarin but greater chance of GIT bleeding
- Works by blocking activity of clotting factor Xa
- Pregnancy category C

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Drugs that affect coagulation

Low molecular weight heparins

- Injected subcutaneously like heparin, but without the major adverse effects
- Safest anticoagulant to use in pregnancy (still a C drug but considered safer than rivaroxaban
- All drugs of this class end in ...parin
  - Enoxaparin (Clexane; know this drug)

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Medications of the Cardiovascular System:
Thrombosis
Lesson 4:
Fibrinolytic drugs

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Fibrinolytic (thrombolytic) drugs

Streptokinase

- Intravenously infused in a hospital setting.
- Potent plasminogen activator
  - Increasing plasminogen increases plasmin activity. Plasmin is the enzyme that dissolves clots.
- Easily results in haemorrhaging (narrow therapeutic range)
- Allergies (of bacterial origin)
- Cannot be re-administered for about 6 months due to allergies developing
  - Alteplase can be used if needed prior to 6 months

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